FPGA-Based Fuzzy Diagnostic System for Heart Rate Monitoring

1st Adrian Widmer, 2nd Alexandre Adonai da Silva, 3rd Eduardo Lôbo Teixeira Filho,
4th Luca Torres Villela, and 5th Natalia Martins Dias Santos

Computer Engineering, SENAI CIMATEC University
Salvador, Brazil
{adrian.widmer, alexandre.silva, eduardo.filho, luca.villela, natalia.santos}@aln.senaicimatec.edu.br

***Abstract*—Cardiovascular monitoring systems typically rely on rigid algorithmic thresholds that fail to account for the inherent physiological variability associated with patient demographics. To address this limitation, this paper elucidates the design and implementation of a hardware-accelerated Fuzzy Inference System (FIS) for real-time heart rate diagnosis on a Field-Programmable Gate Array (FPGA). The proposed architecture integrates a deterministic signal processing pipeline—comprising ROM-based ECG injection derived from a calibrated dataset of 80 patients, Finite State Machine (FSM) QRS complex detection, and a parallel fuzzy classifier. A novel contribution of this work is the concurrent stratification of diagnostic outputs across three distinct age groups (Child, Adult, and Elderly), allowing for context-aware interpretation of vital signs at the sensor edge. Synthesized on a Cyclone V FPGA, the design demonstrates exceptional efficiency, occupying only 3% of adaptive logic modules (ALMs) while maintaining strict timing constraints. Experimental validation confirms the system's ability to mitigate false negatives by correctly distinguishing pediatric bradycardia from normative adult heart rates, establishing a robust framework for low-power, safety-critical biomedical wearables.**

Keywords: FPGA, Fuzzy Logic, QRS Detection, Heart Rate Monitoring, VHDL, Biomedical Edge Computing.

I. Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally, presenting a critical challenge for public health systems. According to the World Health Organization (WHO), early detection and continuous monitoring of cardiac parameters are essential for reducing mortality rates and improving patient prognosis [1]. Among non-invasive diagnostic methods, the Electrocardiogram (ECG) stands as the gold standard for analyzing the heart's electrical activity, allowing for the identification of anomalies through morphological analysis, particularly the QRS complex [2].

However, interpreting ECG signals involves significant complexity due to physiological variability. Parameters such as the normal range of Beats Per Minute (BPM) fluctuate considerably with age and physical condition. Traditional algorithmic approaches based on rigid thresholds (crisp logic) often fail to capture this biological uncertainty, leading to potential misdiagnoses [3]. Consequently, Fuzzy Logic has emerged as a vital theoretical framework in biomedical engineering, providing a mathematical formalism that mimics the nuanced nature of human medical reasoning [4].

Despite these algorithmic advances, the demand for wearable, real-time monitoring devices necessitates architectures that balance computational complexity with power efficiency. Field-Programmable Gate Arrays (FPGAs) offer a distinct advantage by supporting highly parallel, deterministic processing, which is critical for safety-critical healthcare applications [5].

Therefore, the central object of study of this research lies in the **embedding of human-like diagnostic logic directly into hardware**. By translating subjective medical assessment into a precise digital circuit, this work aims to bridge the gap between biological uncertainty and the rigid processing of FPGA architectures.

In this context, this paper presents the implementation of an FPGA-based fuzzy diagnostic system. The proposed architecture includes a VHDL-described signal processing pipeline featuring ROM-based ECG injection, FSM-based QRS detection, and a Mamdani Fuzzy Inference System (FIS). A key contribution is the stratification of diagnosis across three age groups (Child, Adult, Elderly), ensuring a personalized classification of heart rate status at the sensor edge.

II. Theoretical Background

The development of hardware-based diagnostic systems requires a multidisciplinary approach, bridging biomedical signal processing, soft computing, and digital system design. This section outlines the fundamental concepts of ECG analysis and Fuzzy Logic, followed by a systematic review of related works in FPGA-based healthcare monitoring.

A. ECG Signal Characteristics

The Electrocardiogram (ECG) reflects the electrical depolarization and repolarization of the heart. The most prominent feature within a cardiac cycle is the **QRS complex**, which corresponds to the depolarization of the ventricles. Its high amplitude and characteristic steep slope differentiate it from other waves (P and T) and noise artifacts [2].

From a signal processing perspective, the accurate detection of the R-peak (the highest point of the QRS complex) is crucial. The time elapsed between two consecutive R-peaks, known as the R-R interval, is the fundamental variable for assessing cardiac rhythm. While software-based approaches often use complex frequency transforms to identify these features, hardware-constrained architectures typically rely on time-domain analysis—detecting amplitude thresholds and slopes—to estimate the interval in real-time with minimal latency.

B. Fuzzy Logic in Medical Diagnosis

Medical diagnosis is inherently non-binary. A heart rate of 95 BPM might be considered "Normal" for a child but "Tachycardia" for a resting adult. Classical logic (crisp sets) imposes rigid thresholds, which often fails to capture the "gray areas" of physiological status.

Fuzzy Logic, introduced by Zadeh [3], manages this uncertainty by allowing partial membership to a set. A Fuzzy Inference System (FIS) typically consists of three stages:

1) **Fuzzification:** Converting crisp numerical inputs into fuzzy linguistic sets (e.g., Low, Normal, High) using membership functions.
2) **Inference Engine:** Applying "IF-THEN" rules derived from medical expertise to determine the output state based on the inputs.
3) **Defuzzification:** Converting the fuzzy result back into a quantifiable output or a distinct diagnostic class.

C. Systematic Literature Review of FPGA-Based Monitoring

A review of recent literature reveals a growing trend in migrating biomedical algorithms from software (microcontrollers/PCs) to hardware (FPGAs/ASICs) to ensure deterministic performance.

Panigrahy *et al.* [9] demonstrated the efficacy of FPGAs in reducing power consumption for wearable ECG monitoring. Their work implemented a QRS detection algorithm on a Xilinx platform, highlighting the speed advantages over microcontroller-based solutions. However, their system focused primarily on signal extraction rather than high-level diagnostic classification.

Building on this, Benbelkacem *et al.* [10] proposed a more robust FPGA architecture for heart rate monitoring using digital filtering to remove baseline wander. While effective in signal conditioning, the interpretation of the heart rate data remained linear, without accounting for patient demographics such as age groups.

More recently, Garcia-Bermudez *et al.* [11] implemented complex Wavelet Transform algorithms on FPGA for precise R-T interval measurement. While this approach offers high accuracy for arrhythmia detection, the computational resource usage (DSP blocks and logic elements) is significantly higher, potentially limiting its use in low-power, edge-computing devices.

Research Gap: The analysis of the state-of-the-art indicates that while efficient QRS detection on FPGA is well-established, there is a lack of architectures that integrate **context-aware diagnostics** directly into the hardware. Most existing systems output a raw BPM value, leaving the interpretation to external software. This work addresses this gap by embedding a multi-age Fuzzy Inference System directly on the chip, enabling autonomous, personalized classification at the sensor edge.

III. Hardware Architecture and Implementation

The proposed architecture for real-time heart rate monitoring and diagnostic classification was fully implemented in VHDL and deployed on a Field-Programmable Gate Array (FPGA). The FPGA platform was selected due to its ability to support deterministic and highly parallel processing, minimizing latency and improving real-time performance—key requirements for biomedical signal analysis.

A. Pre-processing and ROM Memory Generation

The input electrocardiogram (ECG) signal, originally provided in CSV format (*ptbdb_normal.csv*), was pre-processed using a Python script. The ECG dataset used in this work was obtained from the Heartbeat Categorization Dataset available on Kaggle [6]. The script selected data from 80 patients, flattened the array, and scaled all values into a 12-bit representation (0–4095). These scaled samples were then converted into a VHDL constant array named `ROM_DATA`, implemented as read-only memory to simulate real-time ECG sampling within the diagnostic circuit.

B. QRS Peak Detection and BPM Calculation

The main processing system operates synchronously at 94 Hz, generated from the main 50 MHz input clock via a frequency divider. Data flow control and heart rate estimation are managed by two Finite State Machines (FSMs):

- **Cycle Control (Moore FSM):** Defines a sampling duty cycle of six reading cycles followed by two idle cycles (6T0 to 2T0).
- **QRS Peak Detection (Mealy FSM):** Analyzes each incoming ECG sample (`sample_data`) to identify QRS peaks and measure the R–R interval. Key states include:
  - `ST_IDLE`: waits for a rising slope indicative of a QRS peak (Sample > 3480);
  - `ST_MEASURING`: increments the internal peak timer (`peak_timer`).

The logic flow for determining the heart rate is visually detailed in Fig. 1. The algorithm includes noise filtering by ignoring peaks with insufficient duration (timer ≤ 5).

Once a valid QRS peak is detected, the heart rate is estimated using:

$$\text{BPM} = \frac{7500}{\texttt{peak\_timer}} \tag{1}$$

The computed result is forwarded as an 8-bit signal (`bpm_in`) to the fuzzy diagnostic stage.

C. Fuzzy Inference Module and Defuzzification

The BPM value is processed by a Mamdani-type Fuzzy Inference System (FIS) designed to classify the heart rate into three categories—Low, Normal, High—across three age groups: Child, Adult, and Elderly. The implementation strategy for the fuzzy module follows methodological concepts and examples discussed in the resource provided by Volganga [7].

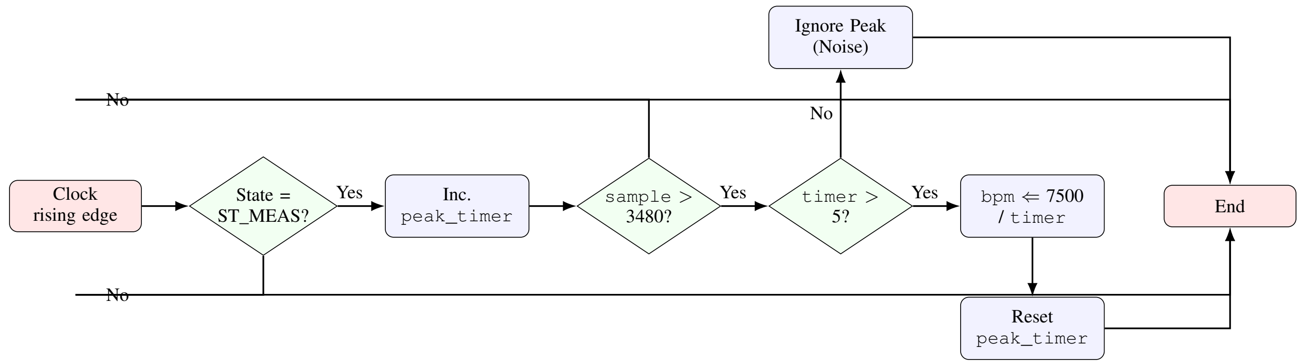

Fig. 1. Flowchart of the logic implemented in the FPGA to determine the heart rate. The system increments a timer during the measuring state and validates the peak against a threshold (3480) and a minimum duration (5 cycles) to filter noise.

In addition, the fuzzy logic model was also prototyped and validated in Python. This software-based implementation facilitated rapid experimentation with membership functions, rule tuning, and verification of the defuzzification strategy before translating the model to VHDL for FPGA deployment.

1) Fuzzification and Membership Functions: Membership functions (μ) were defined using triangular, rising-trapezoidal, and falling-trapezoidal shapes. These functions map the crisp BPM input into membership degrees in the range 0–255. Scaling factors (e.g., $\times 10$ and $\times 25$) were applied to ensure adequate output dynamic range, as follows in Fig. 2.

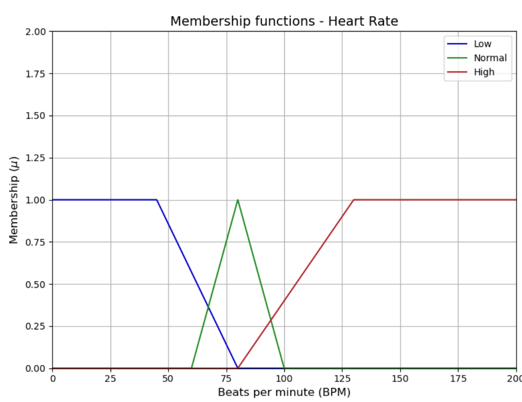

Fig. 2. Membership functions - Heart Rate.

2) Defuzzification: Defuzzification is performed via the *Winner Takes All* method. The final diagnostic output (`diag_child`, `diag_adult`, `diag_elder`) for each age range is determined by selecting the category with the highest membership value.

IV. Results and Discussion

The proposed hardware architecture was synthesized and implemented on the Cyclone V FPGA (Device 5CEBA4F23C7)[8]. The validation process involved both Register Transfer Level (RTL) simulation and physical deployment using the DE0-CV development kit.

A. Resource Utilization

One of the primary objectives of this work was to achieve a lightweight design suitable for wearable applications. Table I summarizes the hardware resources consumed by the system. The complete design utilizes only 3% of the available Adaptive Logic Modules (ALMs) and 6% of the Block Memory bits [8]. This low resource occupancy confirms the system's efficiency and indicates significant available capacity for future integrations, such as wireless communication modules or more complex arrhythmia detection algorithms.

TABLE I
FPGA Resource Utilization (Cyclone V)

Resource	Used	Total	Percentage
Logic Utilization (ALMs)	564	18,480	3%
Total Registers	110	–	–
Total Block Memory Bits	196,608	3,153,920	6%
Total DSP Blocks	6	66	9%
Total Pins	47	224	21%

B. Real-Time Processing and QRS Detection

The system successfully processed the ECG signal injected via the ROM1. The pre-processing step selected data from 80 patients from the CSV file. The Finite State Machine (FSM) responsible for QRS detection operated synchronously at 94 Hz 3, identifying peaks based on the slope threshold (Sample $>$ 3480)4

Fig. 3 illustrates the physical validation of the system. The 7-segment display on the left (01) indicates the current patient ID being processed from the dataset. The display on the right (072) shows the calculated Heart Rate in real-time, which is 72 Beats Per Minute (BPM). This value serves as the crisp input (x) for the subsequent fuzzy inference stage.

C. Fuzzy Diagnostic Classification

The core contribution of this system is the multi-age fuzzy classification. Unlike binary comparators that use a single threshold, the Mamdani inference engine evaluates the 72 BPM input against membership functions for three distinct demographic profiles: Child, Adult, and Elderly (Idoso).

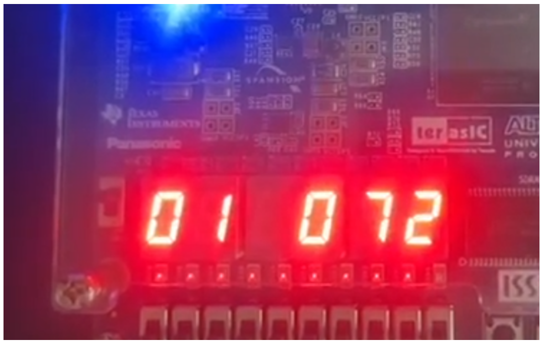

Fig. 3. FPGA hardware implementation showing the current Patient ID (01) and the calculated Heart Rate (072 BPM).

Fig. 4 demonstrates the parallel diagnostic output. The display segments correspond to the linguistic variables assigned to each group:

- **C L (Child → Low):** For a pediatric profile, a heart rate of 72 BPM is classified as "Low" (Bradycardia), as children typically require higher resting rates.
- **A n (Adult → Normal):** For an adult, 72 BPM falls well within the "Normal" trapezoidal membership function (50–100 BPM).
- **I n (Elderly → Normal):** For the elderly profile (denoted by 'I' for *Idoso*), the rate is also classified as "Normal".

Fig. 4. Simultaneous fuzzy diagnostic output. The display indicates: Child = Low (CL), Adult = Normal (An), and Elderly = Normal (In) for the same 72 BPM input.

The simultaneous display of these differing diagnoses for a single input value (72 BPM) validates the system's ability to contextualize physiological data. While a simple threshold algorithm might classify 72 BPM universally as "Normal," the FPGA-based fuzzy logic correctly identifies the potential anomaly for the pediatric case.

D. Discussion

The implementation results highlight the advantages of hardware-based fuzzy logic over traditional software approaches. The parallel nature of the FPGA allowed the three diagnostic inferences (Child, Adult, Elderly) to occur concurrently within the same clock cycles, ensuring deterministic latency. Furthermore, the utilization of integer arithmetic for fuzzy membership calculations (scaling factors of 10 and 25) eliminated the need for floating-point units, contributing to the minimal ALM usage observed in Table I.

This architecture bridges the gap between biological uncertainty and digital precision. By embedding expert medical knowledge into the circuit—specifically the nuances between age groups—the system provides a more robust initial assessment at the sensor edge, reducing the risk of false negatives in vulnerable populations like children [9].

V. Conclusion

This work successfully demonstrated the implementation of a real-time, fuzzy logic-based diagnostic system for heart rate monitoring on an FPGA platform. By translating subjective medical reasoning into a precise digital circuit, the system addressed the challenge of biological uncertainty, providing distinct diagnostic outputs for Child, Adult, and Elderly profiles simultaneously.

The experimental results obtained from the physical deployment on the DE0-CV kit validated the architecture's robustness. The system effectively processed ECG signals from 80 patients, accurately detecting QRS peaks and converting the derived 72 BPM input into context-aware diagnoses (e.g., classifying it as "Low" for children while "Normal" for adults and the elderly).

Furthermore, the synthesis report highlighted the efficiency of the proposed design. With a resource occupancy of only 3% of the available ALMs and 6% of memory bits [8], the architecture proves to be highly scalable. This low utilization suggests significant headroom for future enhancements, such as the integration of wireless communication protocols or more complex Heart Rate Variability (HRV) analysis algorithms, without the need for hardware replacement [11].

In conclusion, this research confirms that FPGA technology, combined with fuzzy logic, offers a powerful alternative to microcontroller-based solutions. It delivers the determinism and parallel processing required for safety-critical healthcare applications while maintaining the flexibility to adapt to diverse patient physiological parameters.

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